

Unit

2

Lesson

3



Innovators in Action.

Wired for Words.

Time Frame: [80-90 min].

Ready, Set, Learn!

By the end of this lesson students will be able to:

- Identify the function of copper tape in conducting electricity and explain how it completes a circuit on fabric.
- Assemble a basic circuit using breadboards and copper tape, following a planned glove layout.
- Understand the role of each Arduino pin connected to the glove components.
- Interpret and write parts of Arduino code that read values from circuit during movement and display output on an LCD.
- Apply logic to connect specific finger movements to printed or displayed messages, using if-then statements in code.

Challenge questions of the lesson.

- What role does copper tape play in making the glove work as a circuit? How does it help electricity flow on fabric?
- How would you organize and connect copper tape and components on the glove to match your planned design? What challenges might you face in assembling this circuit?
- Why is it important to connect each finger's copper tape to specific Arduino pins? What could happen if pins are connected incorrectly?
- How does the Arduino read data from the glove's circuit during finger movements, and how does it display messages on the LCD?
- What logic do you use in if-then statements to ensure the Arduino shows the right message for each finger touch? How would you write these conditions in code?

SEL Relation

- **Self-Management:** Encouraging students to persist through the iterative process of circuit building and coding, managing frustration and celebrating small successes.
- **Social Awareness:** Promoting empathy and understanding as students design technology to assist people with communication challenges, recognizing diverse needs.
- **Relationship Skills:** Supporting collaboration during hands-on activities and peer troubleshooting, building communication and teamwork skills.
- **Responsible Decision-Making:** Developing problem-solving skills as students plan, test, and refine their glove prototypes responsibly and thoughtfully.

Engage



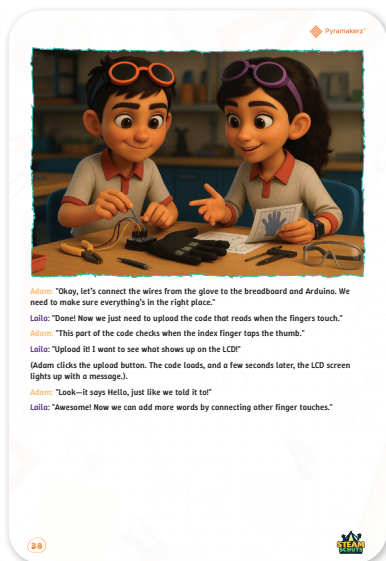
Teacher's Role:

- Read the story aloud clearly and with enthusiasm, emphasizing the connection between the glove, circuit, and code.
- Use gestures or visual aids to help students visualize the wiring, code uploading, and the LCD screen message.
- Pause to check for understanding and invite questions about how the circuit and code interact.
- Facilitate the “Let’s Think” activity by reading the question aloud and encouraging students to brainstorm ideas.
- Provide support to students who may struggle with the technical language or concepts.
- Encourage all students to participate and share their ideas.

Expected from students:

- Listen actively to the story, visualizing how the glove, circuit, and code work together.
- Ask relevant questions to clarify understanding.
- Reflect on the “Let’s Think” question and write or discuss their ideas on how the system produces messages on the screen.
- Respectfully listen to classmates’ ideas and contribute thoughtfully.

Parts of the book included.



Let's Think!



What needs to happen between our glove, the circuit, and the code to make a message appear on the screen?

Story



Objective:

Gather the students and set a curious tone by saying:

“Today, we’re going to see how our smart glove connects with a circuit and code to send messages to a screen. Let’s follow Adam and Laila as they bring their invention to life.”

- Read the story aloud with expression, pausing at key points to emphasize the connection between hardware (wires, breadboard, Arduino) and software (code, LCD screen).
- Use simple analogies to explain complex ideas, e.g., “The code is like instructions that tell the glove what message to show when a finger touches the thumb.”
- Ask guiding questions during the reading, such as:
 - “Why do you think the code needs to check when the finger taps the thumb?”
 - “What do you expect to see on the screen after uploading the code?”
- Invite students to visualize themselves performing the steps of wiring and coding.
- Encourage brief student predictions about what might happen next or how the glove can be improved.

Possible Strategies for Implementation:

1. Set the Stage with Context

- Begin by briefly reviewing the key components involved: glove sensors, wiring, Arduino, and LCD screen.
- Explain that today’s story shows how these parts work together to display messages.

2. Expressive Story Reading

- Read the story aloud with varied tone and pacing to maintain interest and highlight technical steps.
- Use gestures or visual props to illustrate wiring and coding actions.

3. Pause for Reflection

- At important points, pause to ask students what they think is happening and why.
- Encourage students to imagine themselves doing the wiring or uploading the code.

4. Use Guided Questions

- Ask questions such as:
 - “What happens when the code checks if the finger taps the thumb?”
 - “Why is uploading code important for the glove to work?”

5. Connect to Prior Knowledge.

- Relate the story to previous lessons on sensors, circuits, and programming logic.

- Reinforce how hardware and software must work together in tech projects.

6. Encourage Student Predictions.

- Invite students to predict what messages might appear when other fingers touch.
- Prompt creative thinking about additional features or improvements.

7. Support Diverse Learners

- Provide vocabulary support for terms like “breadboard,” “upload,” “LCD.”
- Use visuals or diagrams to complement the story’s text.

Story Reflection



Objective:

Students will analyze and explain the necessary interaction between the smart glove, circuit, and code to generate messages on an LCD screen, reinforcing systems thinking and integration concepts.

Answers:

- “The glove detects when a finger touches the thumb, sending an electrical signal through the wires.”
- “The circuit carries the signal to the Arduino, which processes the input.”
- “The code tells the Arduino what message to display on the LCD screen when it receives the signal.”
- “All three parts must work together so that the touch is detected, processed, and shown as a message.”
- “Without the code, the glove and circuit wouldn’t know what message to show.”
- “Without the circuit, the signal couldn’t travel from the glove to the Arduino.”
- “Without the glove sensing the touch, there wouldn’t be any input to send.”

Possible Strategies for Implementation:

1. Prompt Individual Reflection

- Ask students to take a moment to think silently about the question.
- Encourage them to write down or sketch their ideas before sharing.

2. Facilitate Pair or Small Group Discussion

- Have students discuss their thoughts with a partner or small group, comparing ideas and building on each other’s understanding.
- Circulate to listen and guide conversations, asking probing questions.

3. Use Guided Questioning

- Prompt deeper analysis by asking:
 - “What role does the glove play in detecting input?”

- “How does the circuit transmit the signal?”
- “What does the code do with the signal to show a message?”
- “Why must all three parts work together?”

4. Encourage Use of Diagrams or Analogies.

- Suggest students draw a simple flowchart or diagram showing how information moves from touch to message.
- Use analogies like a relay race or messenger to explain signal flow.

5. Summarize and Share

- Invite volunteers to share their explanations with the class.
- Highlight common themes and clarify misconceptions.

6. Support Diverse Learners

- Provide sentence starters such as: “The glove, the circuit, and the code”
- Allow verbal, written, or visual responses.

Explorer's Toolkit



Teacher's Role:

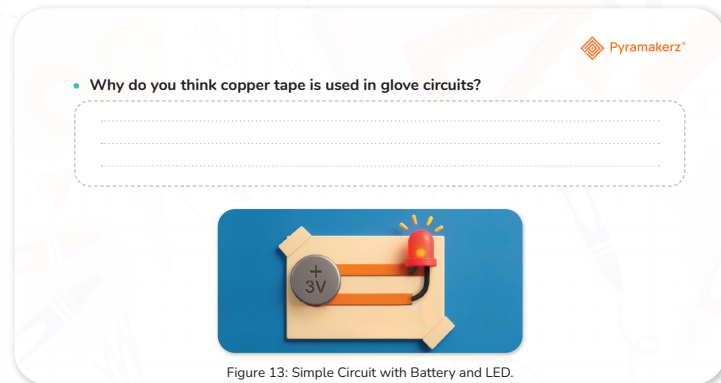
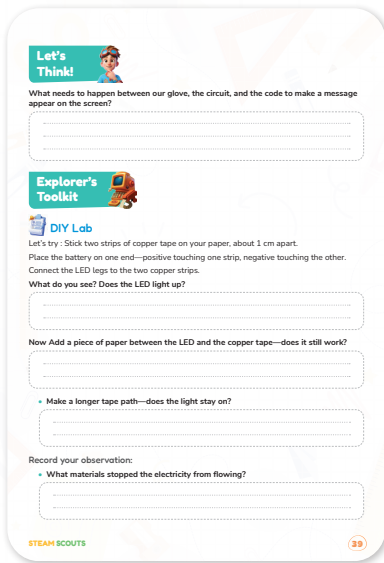
- Prepare materials in advance: strips of copper tape, batteries, LEDs, paper, and scissors.
- Introduce the activity by explaining the goal: to observe how copper tape conducts electricity and affects LED lighting.
- Demonstrate how to place copper tape on paper and connect the battery and LED correctly.
- Supervise students as they build their circuits, offering support with placement and connections.
- Prompt students to make observations and encourage them to test variations like inserting paper between strips or extending tape length.
- Facilitate discussion around reflection questions, guiding students to think critically about their observations.
- Ensure safe handling of batteries and components.

Expected from students:

- Follow instructions carefully to construct the simple circuit on paper using copper tape, battery, and LED.
- Observe and record whether the LED lights up under different conditions.
- Experiment with changing circuit components, such as adding paper or lengthening tape.

- Think critically about why copper tape is used and what materials stop electricity flow.
- Participate actively in group discussions reflecting on the activity.

Parts of the book included.



DIY Lab

Objective:

Students will explore the conductive properties of copper tape by building a simple circuit to light an LED, developing hands-on understanding of electricity flow and materials that conduct or block it.

Answers:

1. What do you see? Does the LED light up?

- The LED lights up when copper tape completes the circuit, allowing electricity to flow.

2. Does it still work with paper between LED and tape?

- No, the LED does not light because paper is an insulator and blocks electrical flow.

3. Does the LED stay on with a longer tape path?

- Usually, yes—though longer paths may cause slight resistance, the LED stays lit if the circuit is complete.

4. What materials stopped the electricity from flowing?

- Non-conductive materials like paper stop the flow by acting as insulators.

5. Why do you think copper tape is used in glove circuits?

- Copper tape is conductive, flexible, and easy to use for creating circuits on gloves.

Activity Instructions:

1. Set Up Circuit Base

- Stick two strips of copper tape on a piece of paper about 1 cm apart.
- Ensure strips are flat and well-adhered.

2. Connect the Battery and LED

- Attach the battery so that the positive end touches one copper strip and the negative end touches the other.
- Connect the LED legs to the two copper strips, noting the correct orientation (long leg to positive).

3. Observe the LED

- Check if the LED lights up when the circuit is complete.
- Record observations.

4. Test with Insulation

- Insert a piece of paper between the LED and copper tape and observe if the LED still lights.
- Record observations on how materials affect electricity flow.

5. Extend the Circuit

- Make one copper tape strip longer and observe whether the LED remains lit.
- Discuss implications for circuit design.

6. Record Observations and Conclusions.

- Answer the reflection questions based on what was seen and learned.

Possible Strategies for Implementation:

1. Pre-Activity Preparation

- Gather and organize materials ahead of time to ensure smooth distribution.
- Review basic concepts of electrical circuits, conductors, and insulators with the class.
- Set clear safety guidelines for handling batteries and LEDs.

2. Demonstration and Modeling

- Model the correct way to place copper tape strips on paper, connect the battery, and attach the LED.
- Highlight the importance of correct LED orientation (long leg = positive).
- Show how to test the circuit and observe the LED lighting.

3. Step-by-Step Guidance

- Guide students through each step, checking for understanding and correct assembly.
- Encourage students to predict what will happen before testing each variation (adding paper, lengthening tape).
- Prompt them to observe carefully and record results.

4. Encourage Inquiry and Experimentation

- Allow students to experiment with different materials to see which conduct electricity and which don't.

- Foster curiosity by asking, “What else do you think might affect the circuit?”

5. Facilitate Group Discussion and Reflection

- After the activity, lead a discussion on their observations and the role of copper tape as a conductor.
- Use guided questions to deepen understanding of circuit concepts and materials science.

6. Support Diverse Learners

- Provide visual instructions or step-by-step checklists for students who benefit from additional structure.
- Offer one-on-one support for students struggling with fine motor skills or concepts.
- Allow alternative ways to document observations, such as drawings or verbal explanations.

7. Reinforce Safety and Responsibility

- Continuously remind students to handle batteries and electrical components safely.
- Emphasize the importance of cleaning up and properly storing materials after the activity.



Teacher's Role:

- Prepare classroom technology for smooth video playback and ensure all students can see and hear clearly.
- Introduce the video by explaining that it will help them learn how different types of signals carry information, which is important for sensors and electronic devices.
- Set a viewing purpose by asking students to pay attention to examples of analog and digital signals and how they differ.
- Pause the video at key points to clarify concepts and check for understanding.
- Facilitate discussion afterward, prompting students to summarize key differences and relate them to their own projects.
- Provide support with vocabulary such as “analog,” “digital,” “continuous,” and “discrete.”

Expected from students:

- Watch the video attentively, focusing on the characteristics of analog and digital signals.
- Take notes or mentally track important concepts and examples.
- Ask questions if something is unclear.
- Participate in post-video discussions by sharing summaries or examples.

- Connect the video content to how their glove sensors might send data.

Parts of the book included.



Screen Lessons

Let's level up our Arduino skills by watching a video that explains the difference between analog and digital signals.

<https://youtu.be/UhNBgdEJGhk?si=uH-dkq8NS81-TwmF>



Screen Lessons

Objective:

Students will understand the difference between analog and digital signals by watching a video explanation, building foundational knowledge relevant to how sensors and circuits communicate data.

Video Link: <https://www.youtube.com/watch?v=UhNBgdEJGhk>



Possible Strategies for Implementation:

1. Pre-Viewing Preparation

- Briefly review prior knowledge about signals or data types.
- Introduce key vocabulary and concepts to look out for.

2. Active Viewing

- Play the video once without interruption to provide a complete overview.
- Replay specific sections that show crucial comparisons or examples.
- Pause occasionally to discuss and answer student questions.

3. Use Visual Supports

- Provide graphic organizers comparing analog and digital signals.
- Use diagrams or real-life examples (like volume knobs vs. on/off switches) to illustrate concepts.

4. Guided Discussion

- Ask students to explain the difference between analog and digital signals in their own words.
- Encourage students to relate this to their project's sensors and how data might be processed.

5. Extension Activity

- Have students brainstorm devices they use daily that rely on analog or digital signals.
- Discuss advantages and limitations of each signal type.

6. Support for Diverse Learners

- Provide vocabulary lists with definitions and visuals.

- Allow students to respond verbally, visually, or in writing to demonstrate understanding.

Elaborate



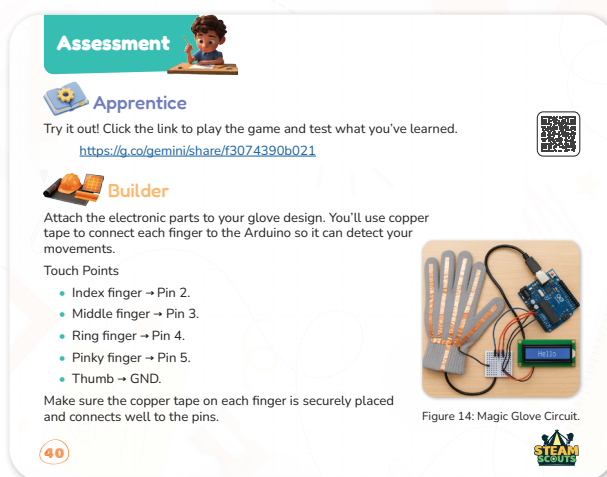
Teacher's Role:

- Provide clear instructions and demonstrations on how to securely attach copper tape from each finger to the correct Arduino pins.
- Monitor students' wiring to ensure accuracy and troubleshoot connection issues.
- Reinforce safety and care when handling electronic components.
- Explain the logic of the if-statements used in the code for detecting finger touches.
- Model how to fill in missing parts of the code, clarifying syntax and semantics.
- Offer guided practice by reviewing students' code snippets and providing feedback.
- Assist students as they upload code to the Arduino and test the glove.
- Encourage systematic debugging by checking wiring, code, and sensor responses.
- Promote peer collaboration for troubleshooting and idea sharing.
- Lead a discussion on what worked well and challenges faced during the build and code process.
- Support students in suggesting improvements and planning next steps.

Expected from students:

- Follow instructions to connect copper tape from fingers to the correct Arduino pins accurately.
- Handle components carefully and safely.
- Understand and complete if-statements to detect finger touches and display corresponding messages.
- Write syntactically correct code and fill in missing conditions as instructed.
- Upload code to the Arduino and observe LCD output.
- Identify any issues and work individually or collaboratively to debug.
- Participate actively in discussions, share experiences, and reflect on their work.
- Be patient and persistent when troubleshooting.

Parts of the book included.



Apprentice

Objective:

Students will demonstrate their understanding of the differences between analog and digital signals by completing an interactive quiz game, reinforcing key concepts in signal processing relevant to sensor technology.

Game Link: <https://g.co/gemini/share/f3074390b021>



Possible Strategies for Implementation:

1. Introduce the Quiz Game

- Explain that the quiz will test their knowledge of analog and digital signals in a fun, interactive way.
- Encourage students to take their time and think carefully about each question.

2. Provide Access and Support

- Ensure all students have access to the game on tablets, computers, or smartphones.
- Assist students who may need help navigating the quiz or understanding questions.

3. Encourage Individual Effort

- Promote independent completion to assess personal understanding.
- Remind students to read questions thoroughly and review their answers.

4. Facilitate Group Discussion Post-Quiz

- After the quiz, discuss common challenges or misconceptions revealed by the questions.
- Clarify any concepts that students found difficult.

5. Use Quiz Results for Feedback

- Review students' performance to identify areas needing review or reteaching.
- Offer personalized guidance based on quiz outcomes.



Objective:

Students will accurately connect copper tape from each finger of their glove prototype to designated Arduino pins, enabling the Arduino to detect finger touches and gather input data, developing hands-on skills in circuit assembly and wearable electronics.

Electronic Components & Circuit Connections:

- The prototype is built using the following electronic components:
 - Arduino Uno
 - LCD 1602 with I2C module.
 - Copper tape (for conductive touch).
 - Jumper wires
 - Glove (any flexible material).

Activity instructions:

1. Prepare Your Materials:

- Gather your glove prototype, copper tape, Arduino board, breadboard (if used), jumper wires, and any other electronic components.
- Make sure your work area is clean and organized.

2. Review Pin Assignments:

- Recall which Arduino pins correspond to each finger:
 - Index finger → Pin 2
 - Middle finger → Pin 3
 - Ring finger → Pin 4
 - Pinky finger → Pin 5
 - Thumb → Ground (GND)

3. Attach Copper Tape on Fingers:

- Carefully place strips of copper tape along each finger on the glove, ensuring they are straight and smooth.
- The tape should start at the fingertip and run down to the glove base where connections will be made.

4. Connect Copper Tape to Arduino Pins:

- Using jumper wires, connect the copper tape strips from each finger to the assigned Arduino digital input pins on the board or breadboard.
- Secure connections to avoid loose wires that can disrupt signals.

5. Connect Thumb to Ground:

- Attach copper tape on the thumb finger to the Arduino ground (GND) pin to complete the circuit for touch detection.

6. Check Connections:

- Double-check all wiring to ensure each finger's copper tape connects to the correct Arduino pin.
- Make sure the copper tape is continuous with no breaks.

7. Test the Circuit:

- Power up the Arduino and test each finger by touching it to the thumb.
- Observe if the corresponding input is detected correctly (e.g., via LED indicators or serial monitor).

8. Troubleshoot Issues:

- If some fingers are not detected, check for loose connections, tape breaks, or incorrect pin assignments.
- Make necessary adjustments.

Possible Strategies for Implementation:

1. Pre-Activity Review

- Begin by reviewing the function of each component: copper tape, Arduino pins, and ground connection.
- Explain the importance of precise connections for proper signal detection.

2. Demonstrate the Assembly Process

- Model how to attach copper tape to fingers and connect them to Arduino pins with jumper wires.
- Highlight the need for secure, continuous electrical paths.

3. Provide Step-by-Step Guidance.

- Walk through each step slowly, ensuring students follow along accurately.
- Check each student's wiring connections individually or in small groups.

4. Encourage Hands-On Exploration.

- Allow students to experiment with taping and wiring, emphasizing trial, error, and adjustment.
- Promote careful observation to identify and fix loose or broken connections.

5. Facilitate Peer Support and Collaboration.

- Encourage students to work in pairs or groups to assist each other with wiring and troubleshooting.
- Foster communication and teamwork during assembly.

6. Use Visual Aids and Diagrams

- Provide clear diagrams showing correct connections between fingers and Arduino pins.
- Use labels on Arduino pins and glove fingers for easy reference.

7. Troubleshoot Proactively

- Teach students basic troubleshooting steps, such as checking continuity of copper tape and correct pin connections.
- Encourage persistence and problem-solving mindset.

8. Reinforce Safety Protocols

- Remind students to handle electronic components carefully and avoid short circuits.
- Monitor for safe use of tools and materials.

Evaluate



Teacher's Role:

- **Code Coach & Clarifier**
 - Review the task prompt with the class: students must write if statements to trigger specific messages when each finger touches the copper tape.
 - Clarify how the `digitalRead()` and `showMessage()` functions work. Provide examples for visual learners and check for understanding.
- **Debugger & Troubleshooting Guide.**
 - Circulate the room and support students in debugging syntax issues or pin connection problems.
- **Prompt students with questions like:**
 - “What pin number is connected to your middle finger?”
 - “What should happen when that pin is LOW?”
- **Reinforcer of Logic Thinking.**
 - Emphasize how the program logic links physical action (finger touch) to a meaningful message.
 - Encourage thoughtful message writing—ask students to match words that might help someone communicate better.

Expected from Students:

- **Write Functional if Statements**
 - Use the given format to write three or more if conditions for different fingers.
 - Assign a clear, purposeful message for each finger (e.g., “Help!”, “Yes”, “No”).
- **Connect Finger Pins Correctly**
 - Match each finger to its assigned pin number and use consistent naming.
 - Apply logic accurately in code blocks to ensure reliable output on the LCD.
- **Test and Debug Their Code**
 - Upload the code and test the glove’s behavior.

- Make adjustments if messages don't display correctly.
- **Reflect on Learning**
 - Review the "Now I Can" section and identify which skills they feel confident in.
 - Update their EDP (Engineering Design Process) worksheet under the "Create" phase to capture what they built and how it works.



Showcase

Objective:

Students will write conditional if-statements to program the Arduino to detect finger touches and display corresponding messages on the LCD screen, demonstrating fundamental coding and hardware integration skills.

Activity instruction:

1. Review Pin Assignments:

- Remind students which Arduino pins correspond to each finger:
 - indexFinger → Pin 2
 - middleFinger → Pin 3
 - ringFinger → Pin 4
 - pinkyFinger → Pin 5

2. Explain LCD Connection Basics:

- Review how the LCD screen is connected to the Arduino (e.g., power, ground, data pins).
- Emphasize that the LCD will display messages based on the code logic.

3. Introduce the If-Statement Format:

- Show the example if-statement:

```
if (digitalRead(indexFinger) == LOW) {  
  showMessage("Hello");  
}
```

- Explain that digitalRead checks if the finger is touched (LOW means touched).
- showMessage displays the text on the LCD.

4. Writing Missing If-Statements:

- Guide students to fill in the missing pins and message texts for the other fingers using the given format.
- Encourage writing clear and correct syntax.

5. Code Upload and Testing:

- Assist students in uploading their complete code to the Arduino.

- Test the glove by touching each finger to verify the correct message appears on the LCD.

6. Debugging:

- If messages do not appear as expected, check wiring, pin assignments, and code syntax.
- Encourage systematic troubleshooting.

7. Iteration and Improvement:

- Discuss how more words or phrases can be added for other finger touches.
- Invite students to think creatively about expanding the code.

Possible Strategies for Implementation:

1. Stepwise Coding Practice:

- Break the coding task into smaller steps and walk through writing one if-statement before moving to others.

2. Visual Code Examples:

- Use a projector or handouts showing correct code snippets to reinforce syntax and logic.

3. Peer Programming:

- Pair students to code collaboratively and review each other's work.

4. Code Debugging Sessions:

- Teach common debugging strategies such as checking pin numbers and matching parentheses/brackets.

5. Use of Commenting:

- Encourage students to write comments in their code to explain each part's function.

6. Provide Coding Templates:

- Give partial code templates where students fill in blanks to reduce overwhelm.

The Code:

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
// Define I2C LCD address and size (usually 0x27)
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
// Finger pin definitions
```

```
const int indexFinger = 2;
```

```
const int middleFinger = 3;
```

```
const int ringFinger = 4;
const int pinkyFinger = 5;

void setup() {
  Serial.begin(9600);

  // LCD setup
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Touch to Speak");

  // Finger pin setup
  pinMode(indexFinger, INPUT_PULLUP);
  pinMode(middleFinger, INPUT_PULLUP);
  pinMode(ringFinger, INPUT_PULLUP);
  pinMode(pinkyFinger, INPUT_PULLUP);
}

void loop() {
  if (digitalRead(indexFinger) == LOW) {
    showMessage("Hello");
  }
  if (digitalRead(middleFinger) == LOW) {
    showMessage("Help");
  }
  if (digitalRead(ringFinger) == LOW) {
    showMessage("Yes");
  }
  if (digitalRead(pinkyFinger) == LOW) {
    showMessage("No");
  }
}

// Function to show message on LCD and Serial
```

```

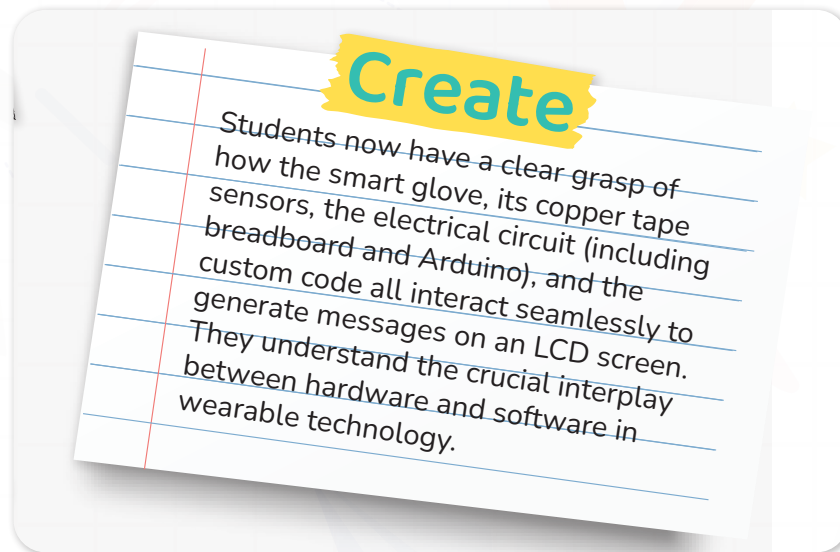
void showMessage(String message) {
  Serial.println(message);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Message:");
  lcd.setCursor(0, 1);
  lcd.print(message);
  delay(1000); // Wait for user to release touch
}

```

Rubric for Coding and Testing Smart Glove

Criteria	Excellent (4 Point)	Good (3 Points)	Fair (2 Points)	Needs Work (1 Point)
Code Accuracy	All if-statements correctly written with proper syntax and pins.	Minor syntax errors or one incorrect pin assignment.	Multiple syntax errors or incorrect pins, but logic mostly correct.	Code incomplete or mostly incorrect syntax/logic.
Functionality	All finger touches correctly display assigned messages on LCD.	Most finger touches display correct messages with minor errors.	Some finger touches display messages; others do not work.	Messages fail to display correctly or no function.
Debugging and Testing	Student identifies and fixes errors independently or with minimal help.	Student fixes most errors with some guidance.	Student requires significant help to debug.	Student unable to debug or test code effectively.
Creativity and Extension	Adds additional messages or features beyond instructions.	Adds minor extensions or custom messages.	Completes basic requirements only.	Does not complete beyond basic requirements.
Presentation and Explanation	Clearly explains code function and logic during demonstration.	Provides reasonable explanation with some gaps.	Explanation is incomplete or unclear.	Unable to explain code or logic.

It's time to move to the Create part and add this.



Now I Can

Objective:

Students will reflect on their learning and identify how they've grown in understanding communication, empathy, and the role of assistive tools in helping others share ideas.

Activity Instructions:

1. Introduce the Reflection

- Say: "Let's look back on everything we've explored this lesson. You learned how it feels to face communication challenges, and how tools can help people share their thoughts in special ways."

2. Guide Students Through the Checklist.

- Read each statement aloud one at a time.
- After each one, ask students to give a thumbs up, sideways, or down to show how confident they feel.
- Example prompts:
 - "Did you learn how people share ideas without talking?"
 - "Do you know how tools can help people talk?"

3. Student Completion

- Let students check off or color the "Now I Can" boxes they feel good about.
- Remind them: "It's okay if you didn't check all of them. We're all still learning."

4. Optional Extension

- Ask students to circle the one they're most proud of.
- Invite them to write a sentence or draw a picture to show their favorite learning moment.

Relation to Standards

1. NGSS (Next Generation Science Standards).

- **MS-ETS1-1:** Define problems that can be solved through engineering design.
- **MS-ETS1-2:** Evaluate competing design solutions based on criteria and constraints.
- **MS-ETS1-3:** Analyze data from tests to determine design improvements.
- **MS-PS2-3:** Apply concepts of forces and motion to understand sensor mechanisms.

2. CCSS (Common Core State Standards).

- **CCSS.ELA-LITERACY.SL.6.1:** Engage effectively in collaborative discussions to solve problems and explain ideas.
- **CCSS.MATH.CONTENT.6.NS.C.6:** Understand and use coordinate planes and integers in mathematical modeling (related to glove mapping and design).

3. ISTE (International Society for Technology in Education).

- **Empowered Learner:** Use technology tools effectively for hands-on projects and coding.
- **Computational Thinker:** Develop algorithmic thinking by writing conditional code for the glove's sensor responses.